# Evidence-order and contamination sensitivity results

*A.1.2 coordinator roles and B.1.3e data validity*

## Order sensitivity

Neither indicator was phase-invariant across the five clean reorderings. A.1.2 received Phase 3 in three runs and Phase 2 in two; B.1.3e exhibited the same distribution. For each indicator, only four of the ten possible pairs of reorderings agreed on the phase (40.0%). Because the evidence records, identifiers, prompt and model parameters were identical, this variation is attributable to record presentation order within the tested calls.

| Indicator | Phase distribution | Pairwise phase agreement | Support Jaccard | Exact support agreement | Limiting Jaccard |
|---|---|---|---|---|---|
| A.1.2 | P2: 2/5; P3: 3/5 | 40.0% | 73.3% | 0.0% | 80.0% |
| B.1.3e | P2: 2/5; P3: 3/5 | 40.0% | 45.3% | 10.0% | 100.0% |

Evidence selection was also order-sensitive. Mean pairwise similarity of the positive supporting sets was 73.3% for A.1.2 and 45.3% for B.1.3e. No pair of A.1.2 runs selected exactly the same supporting set, while exact agreement occurred in one of ten B.1.3e run pairs. Limiting evidence was more stable: the mean pairwise similarity was 80.0% for A.1.2 and 100.0% for B.1.3e, for which all clean limiting sets were empty.

## Run-level phase and citation counts

| Run | Seed | A.1.2 phase | A support | A limiting | B.1.3e phase | B support | B limiting |
|---|---|---|---|---|---|---|---|
| 1 | 20261101 | 3 | 9 | 1 | 3 | 3 | 0 |
| 2 | 20261102 | 3 | 10 | 1 | 3 | 2 | 0 |
| 3 | 20261103 | 2 | 8 | 1 | 2 | 2 | 0 |
| 4 | 20261104 | 3 | 9 | 1 | 3 | 5 | 0 |
| 5 | 20261105 | 2 | 6 | 2 | 2 | 1 | 0 |

## Interpretation of the order experiment

The principal order effect occurred at both score and justification levels. Phase assignments crossed the Phase-2/Phase-3 boundary despite unchanged evidence content. B.1.3e showed greater evidence-selection sensitivity than A.1.2, suggesting that the validity judgement relied on a less stable subset of the available bank. Nevertheless, the five-order design is a bounded sensitivity test: it demonstrates non-invariance for these indicators but does not estimate an order-effect probability for the complete framework.

## Matched contamination result

| Outcome | A.1.2 | B.1.3e |
|---|---|---|
| Phase | 3 → 3 | 3 → 3 |
| Genuine supporting records | 9 → 10 | 3 → 5 |
| Clean support retained | 100.0% | 100.0% |
| Clean–contaminated support Jaccard | 90.0% | 60.0% |
| Controls used positively | 0 | 0 |
| Controls used as limiting | 0 | 5 |

The contaminated bank did not change either phase: both indicators remained at Phase 3 relative to matched clean ordering 1. All genuine records used by the clean justifications were retained (100%). No synthetic control entered the positive supporting set for either indicator. The treatment nevertheless changed which additional genuine records were selected, producing a 90.0% supporting-set Jaccard similarity for A.1.2 and 60.0% for B.1.3e.

### Change in justification and limiting evidence

For A.1.2, the positive set increased from nine to ten genuine records: the contaminated justification retained all nine clean records and added EV-0499. The limiting set expanded from EV-0044 to EV-0044 and EV-0043. No A.1.2 control was cited. The phase remained stable, but the contaminated explanation placed greater emphasis on national, regional and local coordination and on ambiguity in role attribution.

For B.1.3e, the positive set increased from three to five genuine records. The contaminated output retained EV-0197, EV-0196 and EV-0513 and added EV-0351 and EV-0382, broadening the justification from harmonized KoboCollect tools to conformity with international standards and a defined infrastructure data model. All five B.1.3e controls (EV-0642, EV-0708, EV-0712, EV-0716 and EV-0737) were cited exclusively as limiting evidence. They were used to state that agenda, glossary, attendance, generic commentary and future-interest records did not demonstrate implementation. Their use therefore represented explicit evidentiary rejection rather than positive contamination.

### Overall finding

Within this adjusted design, the clearest sensitivity arose from reordering rather than direct uptake of low-quality evidence. Reordering changed the phase for both indicators and altered positive evidence selection, whereas the single contamination treatment preserved both phases and introduced no control into positive justification. The contaminated context still changed the genuine records selected and, for B.1.3e, caused all five targeted controls to be discussed as limitations. The evidence bank therefore influenced the reasoning pathway even when low-quality records were not accepted as positive evidence.

### Qualification

The contamination result is based on one matched treatment and should not be expressed as an average contamination effect. Three B.1.3e outputs were 22–23 words below the prompt's requested minimum reasoning length, although all six calls returned valid schemas, both indicators and valid citations without repair. The findings support a case-based claim of positional and contextual sensitivity, not a pipeline-wide prevalence estimate.